

PHAM MINH HIEU

Robotics Software Engineer | ROS2 | AI Integration | Simulation

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Profile

Senior Robotics Software Engineer with **7+ years** building production-grade autonomous systems. Deep expertise in **ROS2 architecture**, multi-robot coordination, real-time motion control, and simulation-to-real deployment pipelines. Hands-on experience with **NVIDIA Isaac Sim** and **Gazebo** for digital twin development. Proven track record operating robots under live-event pressure at international exhibitions (LogiMat 2026, Sibiu 2025). Passionate about pushing the boundaries of intelligent robotics — from swarm choreography to humanoid motion planning.

Work Experience

Aumovio Singapore

June 2023 – Present

Robotics Software Engineer

- Architected a **scalable multi-robot platform** using **ros2_control** + **ros2_canopen** on Linux embedded systems, achieving real-time control across large fleets of sensors and actuators.
- Developed custom **ros2_controllers** for multiple kinematic configurations (differential drive, omnidirectional, Ackermann) validated in both simulation and physical hardware.
- Built and tested robot behaviours in **NVIDIA Isaac Sim** and **Gazebo Ignition**, establishing reusable digital twin environments for accelerated development.
- Designed and maintained high-level robot control logic with **BehaviorTree-cpp** and ROS2, enabling complex multi-step task execution and autonomous decision-making.
- Fine-tuned **MPPI controller** navigation parameters, increasing operational robot speed to **1.2 m/s** in production environments.
- Deployed and maintained containerised software (**Docker**) across multiple robot platforms, with automated **Jenkins CI/CD** and hardware-in-the-loop / software-in-the-loop test suites.
- **Operated robots on-site** at international events and delivered successful PoCs: *LogiMat March 2026* (Germany) and *Sibiu November 2025* (Romania), troubleshooting live deployments under time pressure.
- Owned VDA5050-compatible system health monitor built on the **ros2 diagnostic_aggregator** framework, enabling real-time fleet monitoring.

Phenikaa X — Phenikaa Group

April 2019 – March 2022

Technical Team Leader, Autonomous Vehicles

- Led R&D of Vietnam's **first Level 4 autonomous vehicle**, integrating open-source stacks (Autoware, Apollo, CommaAI) with custom perception and planning modules.
- Performed full-stack **multi-sensor fusion** calibration: LiDAR, camera, radar, IMU, GNSS, and wheel odometry — directly applicable to quadruped / humanoid sensor integration.
- Ported and optimised the **Autoware autonomous driving stack on LGSVL simulator**, validating autonomous behaviours in virtual test scenarios before real-world deployment.
- Generated **HD maps** for Phenikaa University campus at centimetre-level accuracy using SLAM.
- Implemented **ROS Navigation Stack (Nav2)** for an indoor 2D SLAM robot supporting concurrent users, waypoint missions, and dynamic obstacle avoidance.
- Converted a standard electric golf cart to a **fully autonomous drive-by-wire vehicle** using Autoware, demonstrating end-to-end autonomy from perception to actuation.
- Supervised autonomous system deployments across multiple outdoor test sites; managed multi-person engineering team.

Recognition: **Employee of the Year 2021** | Successfully deployed Autoware on a production minibus — Vietnam's first Level 4 AV in real traffic.

Veriserve Vietnam

April 2022 – May 2023

Senior Software Engineer

- Applied **computer vision** (OpenCV, CNN-based models) across multiple ADAS / autonomous vehicle projects.
- Built a comprehensive **vehicle simulation environment** supporting multiple sensor configurations and driving scenarios for autonomous validation.
- Developed routing algorithms for Model-Based Testing, enabling comprehensive test-case generation from UML State Machines.

FTI — FPT

June 2018 – April 2019

Software Engineer, Robotics Platforms

- Developed base hardware-software solution on a mini electric-car competition platform with integrated **Ubuntu / ROS** stack.
- Built a **real-time automated scoring system** processing telemetry from concurrent autonomous vehicles during live competition events.
- Created a **physics-based simulation** (Unity) for elimination-round competitions.

Recognition: Employee of the Year 2020

Selected Projects Relevant to JTR

Multi-Robot Fleet (ROS2)	Designed ros2_control architecture for simultaneous control of diff-drive, omni, and Ackermann robots; event-tested at trade shows in Europe.
Digital Twin Environments	Built reusable simulation scenes in NVIDIA Isaac Sim and Gazebo for controller development and HIL/SIL testing pipelines.
Behaviour-Based Navigation	Integrated BehaviorTree-cpp with ROS2 actions/services to implement adaptive task execution and multi-step autonomous decision-making.
Autonomous Level-4 Vehicle	End-to-end AV deployment using Autoware on a production minibus — multi-sensor fusion, HD mapping, and real-world traffic operation.
Indoor SLAM Robot	Nav2-based indoor navigation robot supporting concurrent user waypoints and dynamic obstacle avoidance.

Technical Skills

ROS2 (Core)	Nodes, topics, services, actions, TF2, ros2_control, ros2_canopen, diagnostic_aggregator, BehaviorTree-cpp
Programming Languages	Python 3 (Advanced), C++ (Intermediate), C#, JavaScript / TypeScript
Simulation & Digital Twin	NVIDIA Isaac Sim, Gazebo Ignition, LGSVL, Carla, URDF / SDF modelling
Computer Vision	OpenCV, CNN-based object detection, depth camera processing, deep learning inference
Autonomous Navigation	Nav2, MoveIt2, Autoware, Apollo, CommaAI, MPPI controller tuning
Sensor Integration	LiDAR, stereo/depth camera, radar, IMU, GNSS — calibration & multi-sensor fusion
DevOps & Deployment	Docker, Podman, Jenkins CI/CD, OTA updates, AWS
Platforms & Tools	Linux / Ubuntu, Git, Jira, Confluence, Agile, Safety Designer
Hardware	PCB assembly, embedded Linux, actuator maintenance, battery management

Education

Bachelor of Engineering — Control & Automation Engineering

Hanoi University of Science and Technology

Oct 2014 – May 2018

Master of Science — Computer Science (completed coursework)

Phenikaa University

Aug 2021 – May 2023